

TILING SQUARES WITH POLYOMINOES

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ABSTRACT. We consider the question of what squares can be tiled with what polyominoes, for all polyominoes up to pentominoes, considering both allowing and disallowing reflections.

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1. INTRODUCTION

Tiling is critical for both spatial design and aesthetic design. Tiling provides the most efficient packing of objects in a restricted amount of space, and using the same shape to tile creates an aesthetic pattern commonly found in modern design.

Via Conway's Criteria, all polyominoes up to hexominoes can tile the infinite two-dimensional plane. However, restricting the space to squares paints a different picture.

2. MONOMINOES

2.1. **1I monomino.** For the sake of completeness, let us start with monominoes. There is exactly one monomino:



FIGURE 1. 1I monomino

Trivially, the 1I monomino can tile any square with positive integer dimensions.

3. DOMINOES

3.1. **2I domino.** There is exactly one domino:



FIGURE 2. 2I domino

Trivially, the 2I domino cannot tile any square with dimensions not divisible by 2, since tiling via dominoes requires the total number of unit squares be divisible by 2. In general, if an m -omino tiles an $n \times n$ square, then $m \mid n^2$.

We can construct a tiling for the case where the dimensions is divisible by 2 by repeating the following structure:

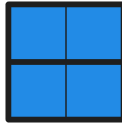


FIGURE 3. 2×2 monomer with 2I domino

Call this a *monomer*. In general, if we can construct an $n \times n$ monomer, we can tile any $kn \times kn$ square. Therefore, the 2I domino can tile an $n \times n$ square iff $2 \mid n$, that is, if n is a multiple of 2.

4. TROMINOES

4.1. **3I tromino.** Let us first consider the 3I tromino:



FIGURE 4. 3I tromino

Following the same logic, since $3 \mid n^2$ is necessary and you can construct a 3×3 monomer via the following, the 3I tromino can tile an $n \times n$ square iff $3 \mid n$.

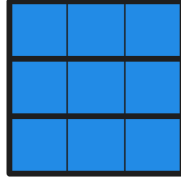


FIGURE 5. 3×3 monomer with 3I tromino

4.2. **3L tromino.** Now let us consider the 3L tromino:

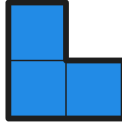


FIGURE 6. 3L tromino

Again, $3 \mid n^2$ is necessary. It is easy to verify that the 3L tromino cannot tile a 3×3 square since, in order to not isolate any cells, the middle cell requires the 3L placement on the right (up to rotation and reflection), which leaves the top right cell impossible to tile.

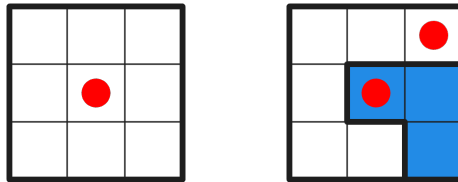


FIGURE 7. The 3L tromino cannot tile a 3×3 square.

Now, we present two monomers: a 2×3 monomer and a 9×9 monomer:



FIGURE 8. 2×3 monomer with 3L tromino

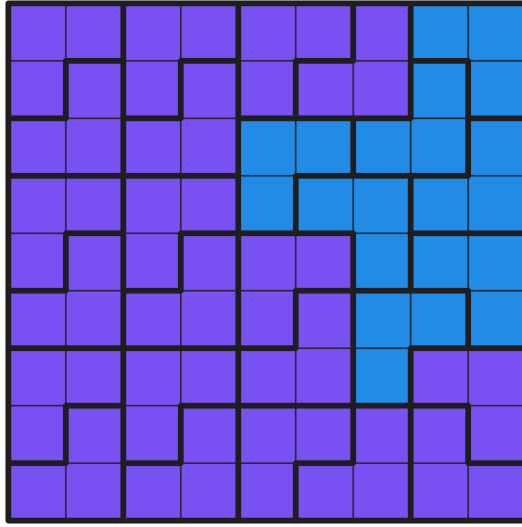


FIGURE 9. 9×9 monomer with 3L tromino

We wish to show how this allows the tiling of all $n \times n$ squares where $3 \mid n$ and $n \geq 6$. First, notice that the 2×3 monomer can repeat to create a 6×6 monomer, thus resolving all cases where $6 \mid n$. Hence, all the cases left are those where $n = 6k + 3$, where k is a positive integer.

We prove that these cases are resolvable via weak induction on k .

Base Case. When $k = 1$, the 9×9 square can be tiled directly with the monomer given in Figure 12.

Induction Hypothesis. Assume that a $(6k + 3) \times (6k + 3)$ square can be tiled for some particular k .

Induction Step. We break a $(6k + 9) \times (6k + 9)$ square into four components: a $(6k + 3) \times (6k + 3)$ square, two $6 \times (6k + 3)$ rectangles, and a 6×6 square. The $(6k + 3) \times (6k + 3)$ square is resolved via Induction Hypothesis, whereas the other three components can be tiled with the 2×3 monomer.

Similarly, whenever we have a $b \times 2b$ monomer and an $ab \times ab$ monomer, we can use the same logic to generate all $(a + 2k)b \times (a + 2k)b$ squares. (The current case is the case where $a = 3$ and $b = 3$, multiplying the 2×3 monomer into a 3×6 monomer.)

Therefore, the 3L tromino can tile an $n \times n$ square iff $3 \mid n$ and $n \geq 6$.

5. TETROMINOES

5.1. **4O tetromino.** Let us first consider the 4O tetromino:



FIGURE 10. 4O tetromino

Since $4 \mid n^2$ is necessary and a 2×2 monomer is trivial, the 4O tetromino can tile an $n \times n$ square iff $2 \mid n$.

Observe how all polyominoes until now are non-chiral, that is, the result remains regardless of whether we allow reflections, since reflections do not provide new shapes.

5.2. **4S tetromino.** We now consider the 4S tetromino.

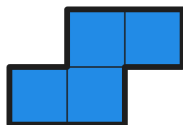


FIGURE 11. 4S tetromino

The 4S tetromino cannot tile any squares, since the corners restrict the 4S tetrominoes in a way that cannot be satisfied:

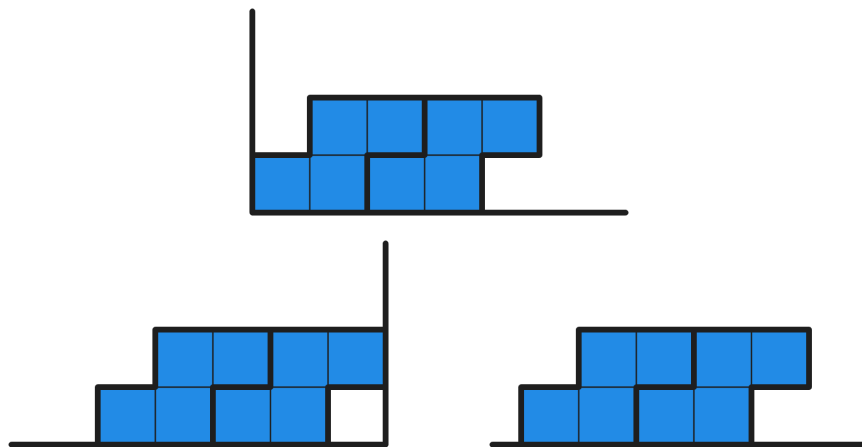


FIGURE 12. The corners force 4S placements into the other corner.

Notice that, even though 4S has chirality, the chirality does not help; regardless of whether we allow reflections, 4S cannot tile any squares.

5.3. **4I tetromino.** Let us now consider the 4I tetromino.



FIGURE 13. 4I tetromino

Clearly, $4 \mid n^2 \iff 2 \mid n$ is a necessary condition, and $4 \mid n$ is a sufficient condition due to the trivial 4×4 monomer:

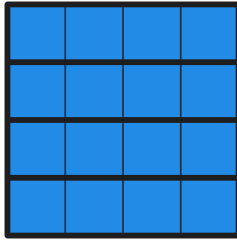


FIGURE 14. 4×4 monomer with 4I tetromino

Now, let us consider the case for squares of size $(4k+2) \times (4k+2)$. We create what is called a coloring argument, by coloring the $(4k+2) \times (4k+2)$ square like the following (specifically, for i from 1 to $4k+2$ and for j from 1 to $4k+2$, color based on $\lfloor (i-1)/2 \rfloor + \lfloor (j-1)/2 \rfloor \pmod{2}$, where 0 corresponds to black and 1 corresponds to white):

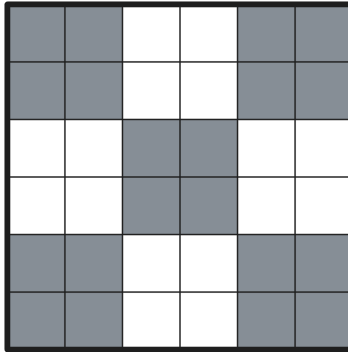


FIGURE 15. Coloring of an 6×6 example.

Recognize how any 4I placement covers 2 black and 2 white cells, but in total there are always 4 more black cells than white squares ($8k^2 + 8k + 4$ compared to $8k^2 + 8k$). Therefore, the 4I tetromino can tile an $n \times n$ square iff $4 \mid n$.

5.4. **4T tetromino.** We now consider the 4T tetromino.

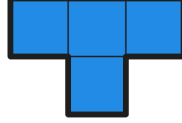


FIGURE 16. 4T tetromino

Clearly, $4 \mid n^2 \iff 2 \mid n$ is a necessary condition, and $4 \mid n$ is a sufficient condition due to the 4×4 monomer:

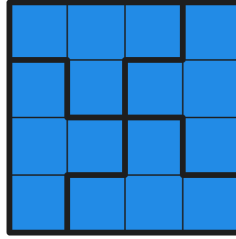


FIGURE 17. 4×4 monomer with 4T tetromino

Now, let us consider the case for squares of size $(4k+2) \times (4k+2)$. We apply a coloring argument based on coloring by $i+j \pmod{2}$:

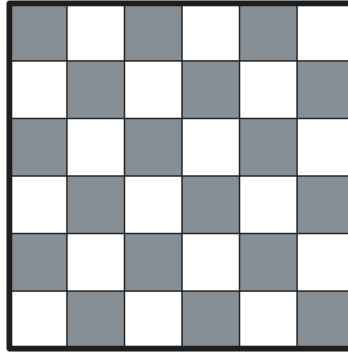


FIGURE 18. Coloring of an 6×6 example.

Recognize how the total number of black and white squares are equal, but any 4T tetromino covers either 2 more black squares than white squares or 2 more white squares than black squares, which cannot average out over a total odd number of 4T tetrominoes (there will always be more black-majority 4T placements or more white-majority 4T placements, leading to more black squares or more white squares). Therefore, the 4T tetromino can tile an $n \times n$ square iff $4 \mid n$.

5.5. **4L tetromino.** We finally consider the 4L tetromino.

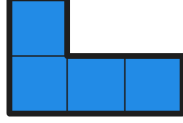


FIGURE 19. 4L tetromino

Clearly, $4 \mid n^2 \iff 2 \mid n$ is a necessary condition, and $4 \mid n$ is a sufficient condition due to the 4×4 monomer (which does not require reflections):

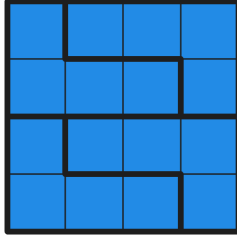


FIGURE 20. 4×4 monomer with 4L tetromino

Now, let us consider the case for squares of size $(4k+2) \times (4k+2)$. We apply a coloring argument based on coloring by $i \pmod{2}$:

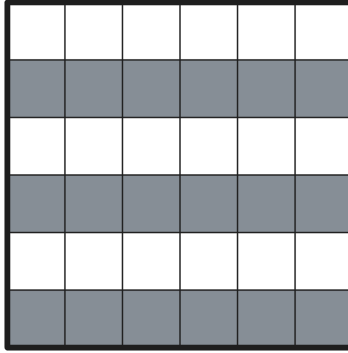


FIGURE 21. Coloring of an 6×6 example.

Recognize how the same argument applies: the total number of black and white squares are equal, but any 4L tetromino covers either 2 more black squares than white squares or 2 more white squares than black squares. Therefore, the 4L tetromino can tile an $n \times n$ square iff $4 \mid n$.

Observe how the impossibility argument holds regardless of whether we allow reflections, and that the construction does not require reflections. Hence, the result holds regardless of whether we allow reflections.

6. PENTOMINOES

6.1. Impossible pentominoes. Several pentominoes trivially cannot tile any square, via reasoning on the corners similar to the argument for 4S:

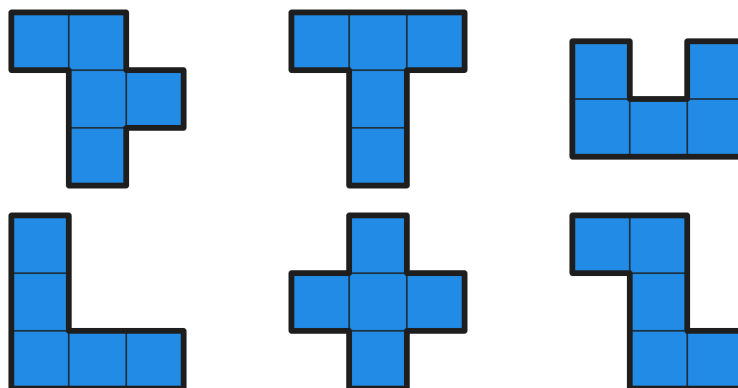


FIGURE 22. Pentominoes which fail covering one corner: in reading order, 5F, 5T, 5U, 5V, 5X, and 5Z.

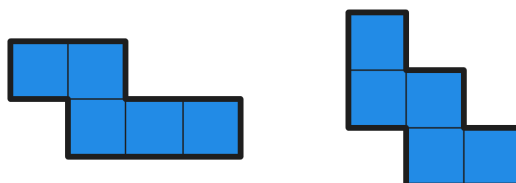


FIGURE 23. Pentominoes which one corner forces into another, creating an impossibility: in reading order, 5N and 5W.

The proof for these cases are mostly rogue casework, so we leave these as an exercise to the reader.

6.2. **5I pentomino.** Let us consider the 5I pentomino:



FIGURE 24. 5I pentomino

Clearly, $5 \mid n^2 \iff 5 \mid n$ is a necessary condition, and $5 \mid n$ is a sufficient condition due to the trivial 5×5 monomer:

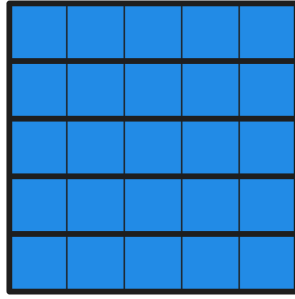


FIGURE 25. 5×5 monomer with 5I pentomino

Therefore, the 5I pentomino can tile an $n \times n$ square iff $5 \mid n$.

6.3. **5L pentomino.** Let us consider the 5L pentomino:

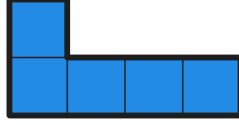


FIGURE 26. 5L pentomino

Again, $5 \mid n^2 \iff 5 \mid n$ is a necessary condition. Observing the 2×5 monomer below, cases where $10 \mid n$ are trivial by multiplying the monomer.

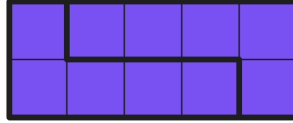


FIGURE 27. 2×5 monomer with 5L pentomino

It is trivial to check that a 5×5 square is impossible to tile. If we allow reflections, then we have the following 15×15 monomer:

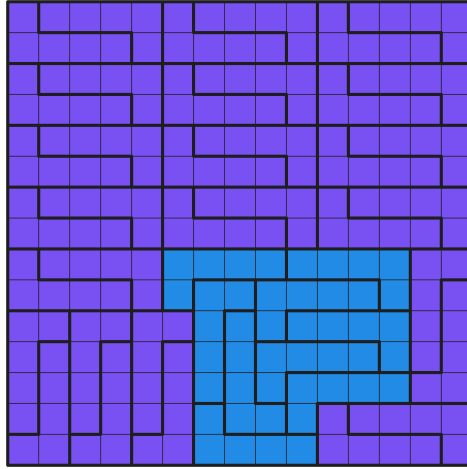


FIGURE 28. 15×15 monomer with 5L pentomino

The same inductive reasoning as the 3L gives that, allowing reflections, the 5L pentomino can tile an $n \times n$ square iff $5 \mid n$ and $n \geq 10$.

If we do not allow reflections, then we have a 25×25 monomer:

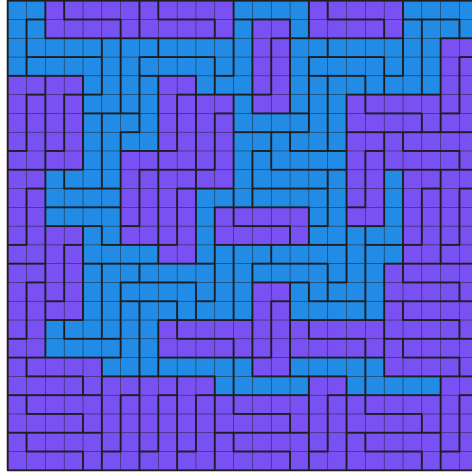


FIGURE 29. 25×25 monomer with 5L pentomino

Using a program, we can verify that a 15×15 tiling does not exist. Hence, disallowing reflections, the 5L pentomino can tile an $n \times n$ square iff $5 \mid n$, $n \geq 10$, and $n \neq 15$. We supply the code below:

```

grid = 0
for _ in range(15):
    grid <= 16
    grid += 2**15 - 1

def pprint(grid):
    for i in range(16):
        for j in range(16):
            print("#"[(grid >> (16 * i + j)) % 2], end=' ')
        print()
    print()
    print()

def lsd(x): return x & -x

def within(grid, pattern): return grid & pattern == pattern

def placements(x):
    return [
        x * 0b1111_0000_0000_0000_1,
        x * 0b1_0000_0000_0000_1111,
        x * 0b1000_0000_0000_0000_1000_0000_0000_1000_0000_0000_11,
        x * 0b11_0000_0000_0000_0001_0000_0000_0001_0000_0000_0000_0001,
    ]

def cache(f):
    d = {}
    def inner(a):
        if a not in d: d[a] = f(a)
        if d[a] == True: pprint(a)
        return d[a]
    return inner

@cache
def possible(grid):
    x = lsd(grid)
    if x == 0: return True
    for p in placements(x):
        if within(grid, p): and possible(grid-p): return True
    return False

print(possible(grid))

```

6.4. **5Y pentomino.** Let us consider the 5Y pentomino:

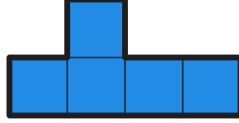


FIGURE 30. 5Y pentomino

$5 \mid n$ is again necessary, and the following 5×10 monomer (which does not require reflections) solves all cases for $10 \mid n$.

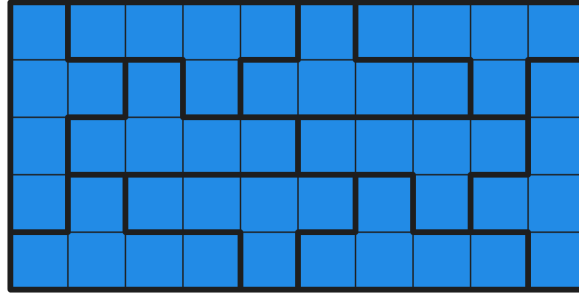


FIGURE 31. 5×10 monomer with 5Y pentomino

It is again trivial to check that a 5×5 square is impossible to tile. If we allow reflections, then we have the following 15×15 monomer:

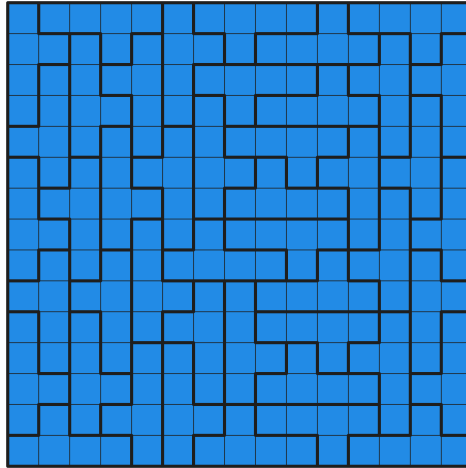


FIGURE 32. 15×15 monomer with 5Y pentomino

Hence, if we allow reflections, the 5Y pentomino can tile an $n \times n$ square iff $5 \mid n$ and $n \geq 10$.

If we do not allow reflections, then it turns out that $10 \mid n$ is necessary. One may attempt a coloring argument, but no coloring argument can work here. Fundamentally, color arguments are impossibility characterizations under modular arithmetic. What this means is that if we consider *signed tilings* — tilings which may also use “negative” polyominoes, i.e. subtracting them from the current configuration — our results should still hold. However, we give a signed tiling of a 1×5 monomer using 5Y pentominoes without reflections:

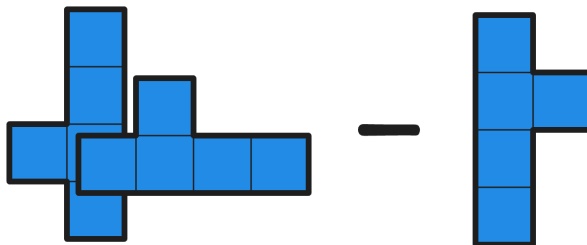


FIGURE 33. 1×5 signed monomer with 5Y pentomino

Therefore, we must rely on another argument. We sketch a beautiful argument by Yura Aksyonov as follows:

Let us take the example of 15×15 . It can be seen through casework that the following two formations are forbidden:



FIGURE 34. Two cave-ins impossible to fill

Therefore, the following placements can be shown to be forced:

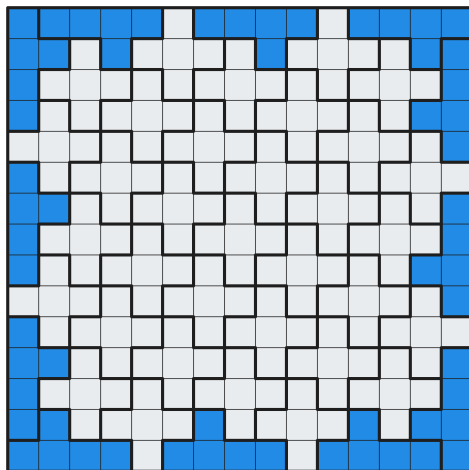


FIGURE 35. Placements forced from the corners to the edges

Notice how the remaining area can be partitioned into $5X$ pentominoes. By considering the marked cell (choosing the most top-right $5X$ among the most top-left slanted line of $5X$ pentominoes), we have only two scenarios:

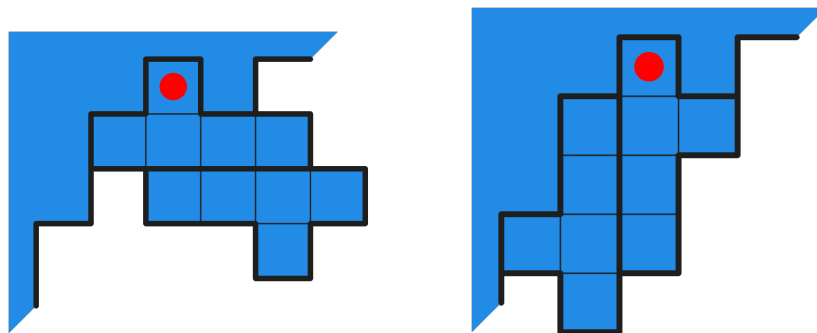


FIGURE 36. The only possibilities of satisfying the marked cell

Realize that, in each case, the two $5Y$ pentominoes cover exactly two $5X$ pentominoes, allowing the procedure to continue. However, we need to place another odd number of $5Y$ pieces (a total odd number of pieces, minus an even number of pieces on the edge), which makes it impossible.

Hence, if we disallow reflections, the $5Y$ pentomino can tile an $n \times n$ square iff $10 \mid n$.

6.5. **5P pentomino.** We finally consider the 5P pentomino.

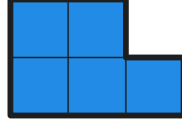


FIGURE 37. 5P pentomino

$5 \mid n^2 \iff 5 \mid n$ is necessary, and we have the 2×5 monomer below yielding cases where $10 \mid n$.



FIGURE 38. 2×5 monomer with 5P pentomino

5×5 is again impossible, but a 15×15 monomer exists:

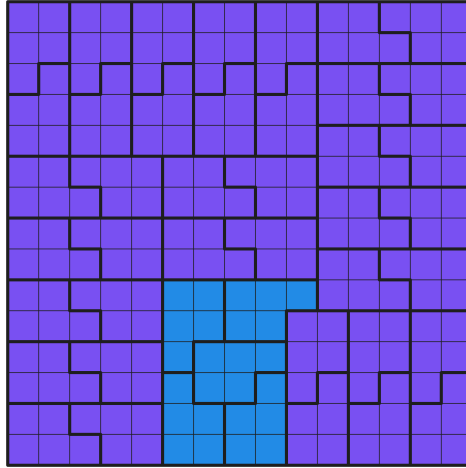


FIGURE 39. 15×15 monomer with 5P pentomino

Hence, if we allow reflections, the 5P pentomino can tile an $n \times n$ square iff $5 \mid n$ and $n \geq 10$.

Again, if we disallow reflections, then $10 \mid n$ is necessary. We first show that no straightforward coloring argument exists with a signed 1×5 tiling:

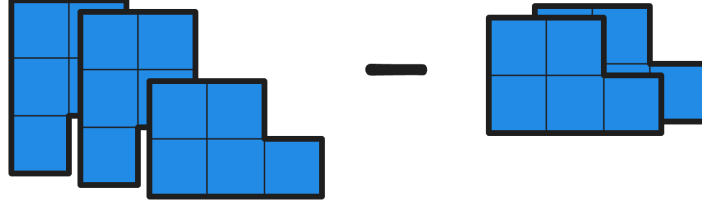


FIGURE 40. 1×5 signed monomer with 5P pentomino

Next, we reproduce a proof given by Michael Reid:

Consider the following permutations in S_{64} :

$x = (1, 2, 4, 47, 16, 27, 41, 54, 56, 9)(3, 6, 12, 11, 34, 50, 62, 61, 49, 58)(5, 10, 19, 32, 24, 36, 31, 37, 42, 55)$
 $(7, 14, 23, 28, 43, 57, 52, 40, 38, 46)(8, 59)(13, 21, 35, 51, 20, 15, 25, 17, 18, 30)(22, 33, 48, 60, 64, 26, 39, 53, 63, 44)(29, 45),$
 $y = (2, 3, 5, 9, 17, 28, 42, 12, 14, 22)(4, 7, 13, 6, 20)(8, 25, 37, 11, 33)(10, 18, 29, 44, 58)(15, 24, 30, 46, 57, 63, 62, 48, 54, 47)$
 $(16, 26, 38, 50, 61)(19, 31, 39, 45, 21, 34, 49, 51, 59, 64)(27, 40)(32, 36, 52, 35, 41)(43, 56, 60, 55, 53)$

It can be validated that $x^3 y x^{-1} y x^{-2} y^{-2} = x^2 y^3 x^{-1} y^{-1} x^{-1} y^{-2} = e$. This is to say that, for any 5P pentomino (with the chirality portrayed above), we can take a walk around that pentomino, performing x when we move right, y when we move up, x^{-1} when we move left, and y^{-1} when we move down, and we will end up with the same permutation since we have performed the equivalent of the identity permutation e . Due to the properties of permutation, we can begin the movement at any point, provided we stop at that same point.

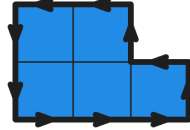


FIGURE 41. One (counterclockwise) cycle around a P5

Notably, this means that, for a tiling of P5, we can traverse each pentomino, wrapping around each pentomino once, again with the net effect of the identity permutation e :

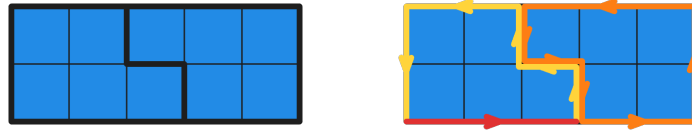


FIGURE 42. An example traversal. We enter from the bottom left to the right, winding counterclockwise around the right pentomino, and then completing the winding around the left pentomino (starting by immediately backtracking the common edges).

However, notice how, during the traversals, the edge in which the two pentominoes touch is traversed twice. If we cancel these traversals (since x cancels with x^{-1} and y cancels with y^{-1}), we are left with the boundary of the entire tiling.

What this is to say is that, if there exists a certain tiling of 5P pentominoes of an $n \times n$ grid, the boundary of the resulting grid, represented by $x^n y^n x^{-n} y^{-n}$, must be e , since we can obtain it by tracing individual 5P tiles of the tiling. However, it is easy to verify that, for all $n = 10k + 5$,

$$x^n y^n x^{-n} y^{-n} = (1, 23)(2, 28)(3, 42)(4, 43)(5, 12)(6, 55)(7, 56)(8, 29)(9, 14)(10, 11)(13, 60)(15, 63)(16, 52)(17, 22)(18, 33)(19, 34)(20, 53)(21, 64)(24, 62)(25, 44)(26, 35)(27, 40)(30, 48)(31, 49)(32, 50)(36, 61)(37, 58)(38, 41)(39, 51)(45, 59)(46, 54)(47, 57) \neq e.$$

Therefore, there must not exist a tiling of 5P pentominoes for $n \times n$ squares when $n = 10k + 5$, meaning that, if we disallow reflections, then the 5P pentomino can tile an $n \times n$ square iff $10 \mid n$.

7. CONCLUSION

In this paper, we have outlined tiling conditions of squares for polyominoes using combinatorial arguments. Many of these conditions are extendable to more shapes, but for the sake of scope, we have limited our discussion to up to pentominoes.

Further research may look into higher-order polyominoes, as well as non-polyomino shapes, or shapes composed of equilateral triangles on a triangular grid tessellating equilateral triangles.

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All visuals were generated using Excalidraw.